E09 Multiple Reactions in an Isothermal PFR

The following reactions are carried out in the gas phase in an isothermal PFR:

$A + 2B \rightarrow C + D$ $r_{1D} = k_{1D}C_AC_B$

$2D + 3A \rightarrow C + E$ $r_{2E} = k_{2E}C_AC_D$

$B + 2C \rightarrow D + F$ $r_{3F} = k_{3F}C_B^2C_C$

Where k_{1D} = 0.25 L/mol*min, k_{2E} = 0.1 L/mol*min, k_{3F} = 5 L^2/mol^2*min, C_{T0} = 0.4 mol/L, V = 500 L, $y_{A0} = y_{B0}$, υ_0 = 100 L/min. Plot the species concentrations as functions of PFR volume.

to get $C_j(V)$, we need to solve the mole balances

$$\frac{dF_j}{dV} = r_{j,net}$$

to get $r_{j,net}$, we use the stoichiometry and rate matrices:

$$r = \begin{bmatrix} k_{1D} C_A C_B \\ k_{2E} C_A C_D \\ k_{3F} C_B^2 C_C \end{bmatrix}$$

reactions are already written s.t. stoichiometric coefficients on D (rxn 1), E (rxn 2), & F (rxn 3) are 1. So, no need to re-write.

$$\nu = \begin{bmatrix} -1 & -2 & 1 & 1 & 0 & 0 \\ -3 & 0 & 1 & -1 & 1 & 0 \\ 0 & -1 & -2 & 1 & 0 & 1 \end{bmatrix}$$